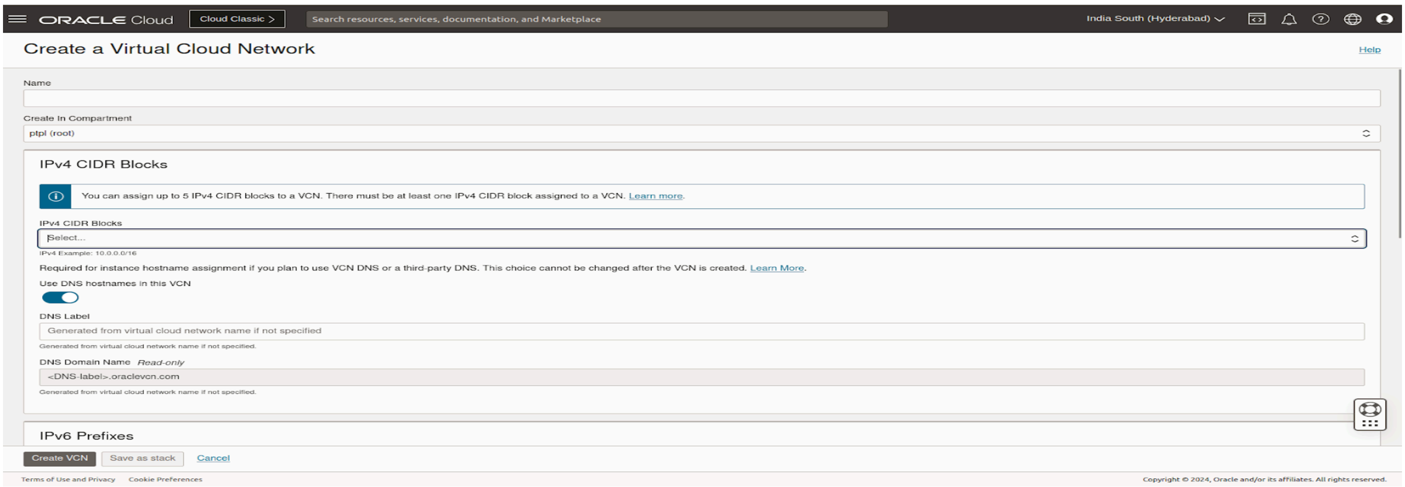
Create two instances in OCI as the backend set's (instance's) to be configured for the load balancer

Creating and Configuring Load Balancer In OCI

1) Login to OCI Console
2) Create a Virtual Cloud Network (VCN)

- Navigate to Networking —> Virtual Cloud Network —> Create VCN
- **Name**: Enter a name for your VCN, e.g., `myVCN`.
- **Compartment**: Select the compartment for your VCN.
- **CIDR Block**: Choose a CIDR block for your VCN, for example, `10.0.0.0/16`.
- **DNS Resolution**: Keep it enabled.
- **DNS Hostnames**: Keep it enabled.
- Keep all other option default and Click **Create**.

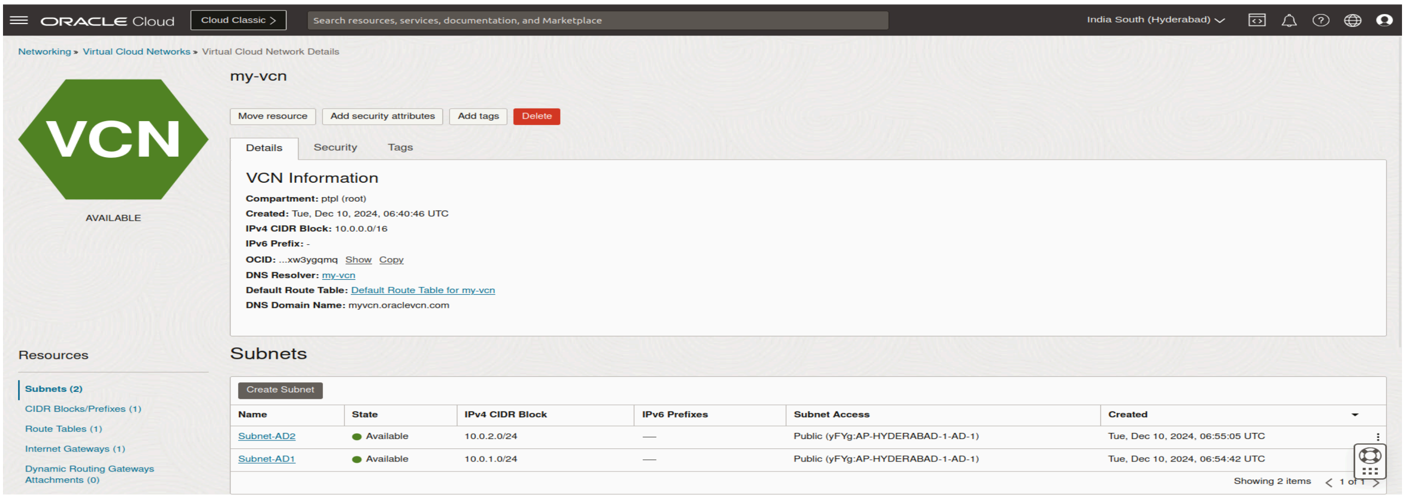


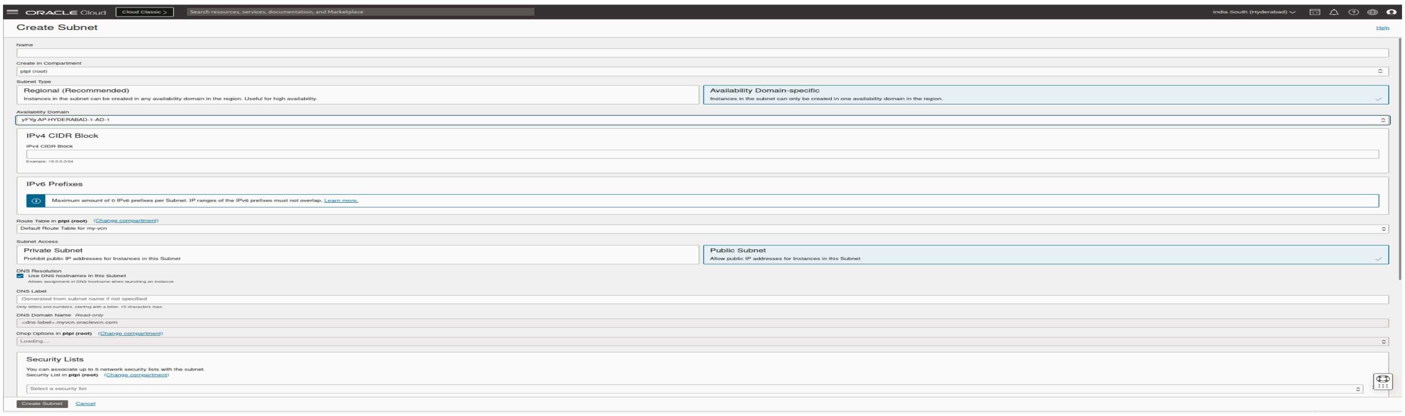
3) **Create Subnets in the VCN**

- After the VCN is created, navigate to the **Subnets** section inside VCN
- Click **Create Subnet**.
- **Name**: Enter a name, e.g., `subnet-AD1`.
- **Compartment**: Select the compartment.
- **CIDR Block**: Enter a subnet CIDR block. For example, `10.0.1.0/24` for the first subnet.
- **Availability Domain (AD)**: Choose the **first AD** (e.g., `AD-1`).
- **Subnet Type**: Select **Public**.
- Keep all other option default and Click **Create**.

Now, create the second subnet:

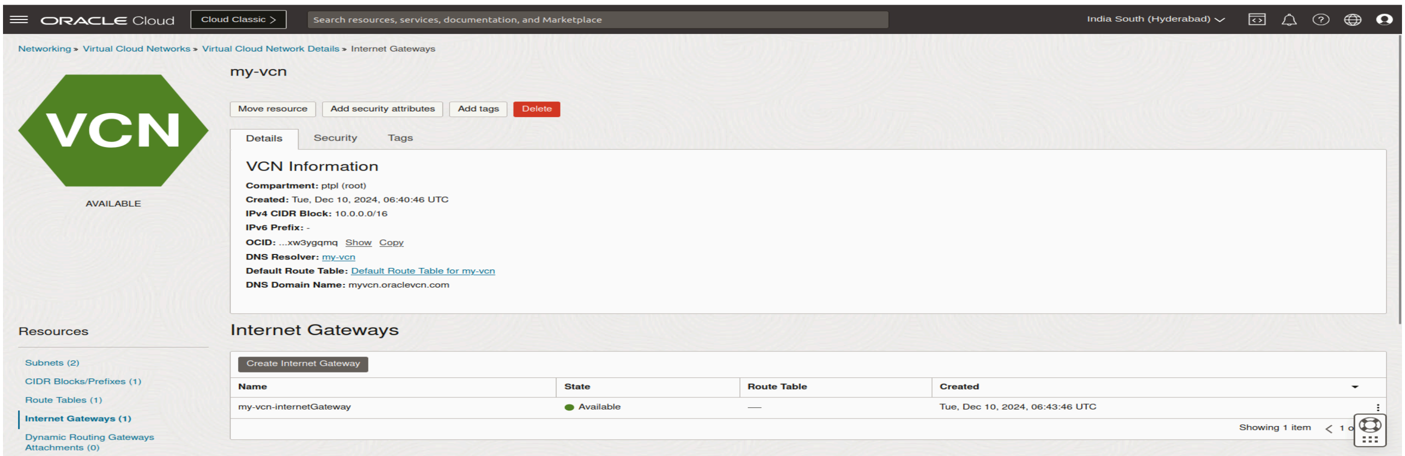
- Click **Create Subnet** again.
- **Name**: `subnet-AD2`.
- **CIDR Block**: Enter another subnet CIDR block, e.g., `10.0.2.0/24`.
- **Availability Domain (AD)**: Choose the **second AD** (e.g., `AD-2`).
- **Subnet Type**: Select **Public**.
- Click **Create**.





4) Create an Internet Gateway (IGW)

- **In the Oracle Cloud Console, under Networking, —> VCN —> My VCN —> click Internet Gateways.**
- **Click Create Internet Gateway.**
- **Name:** Enter a name, e.g., `myIGW`.
- **Compartment:** Choose the correct compartment.
- **Attach to VCN:** Select the VCN you just created (`myVCN`).
- Keep all other option default and Click **Create**.

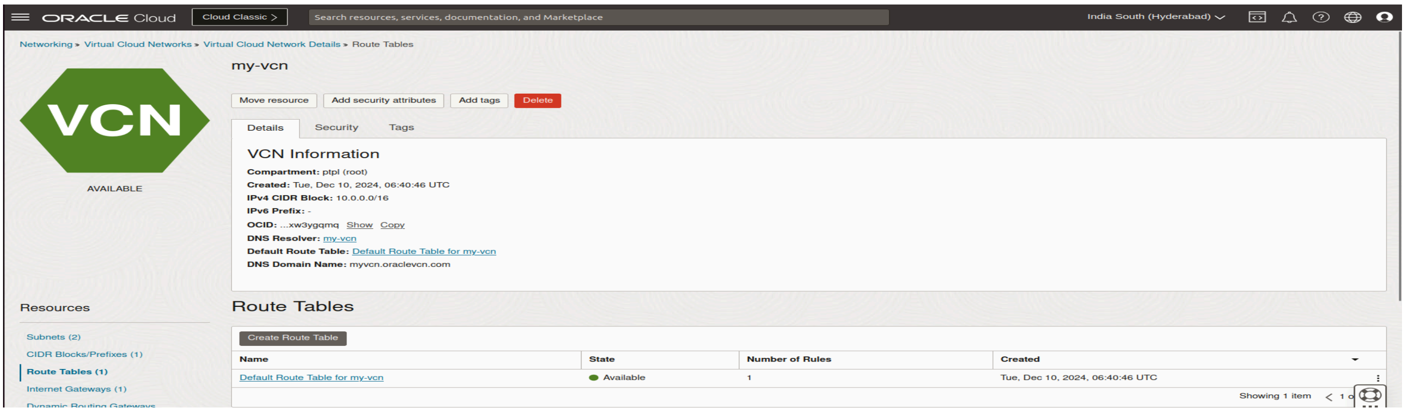


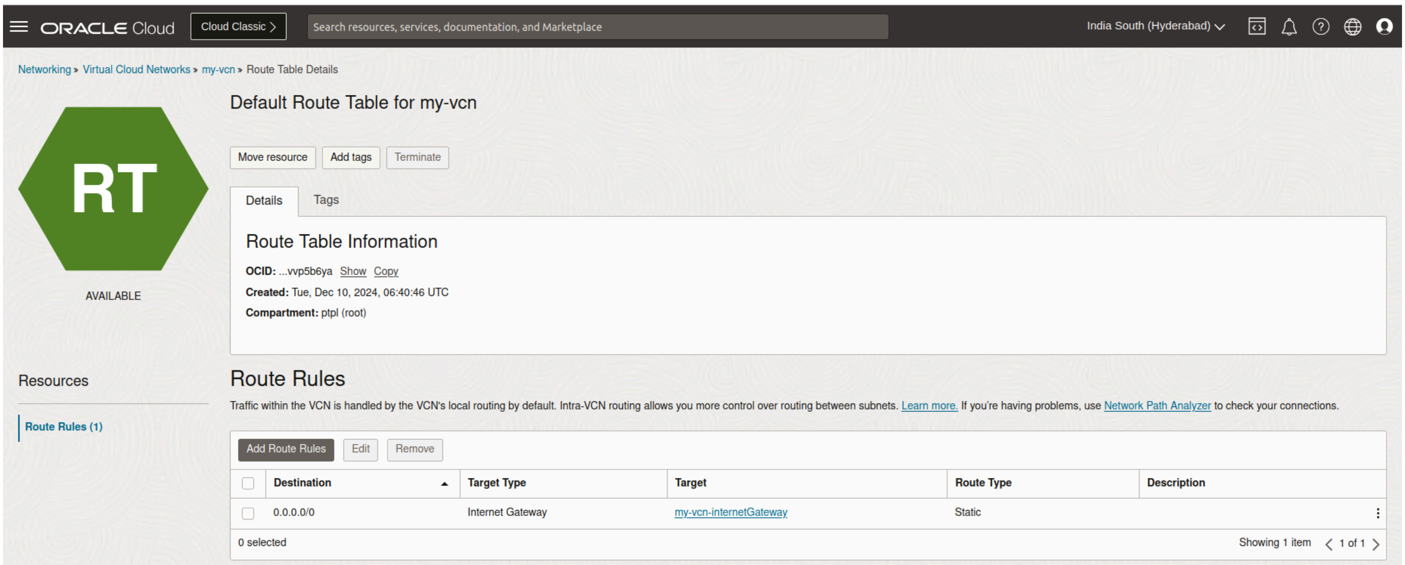
5) Update Route Tables

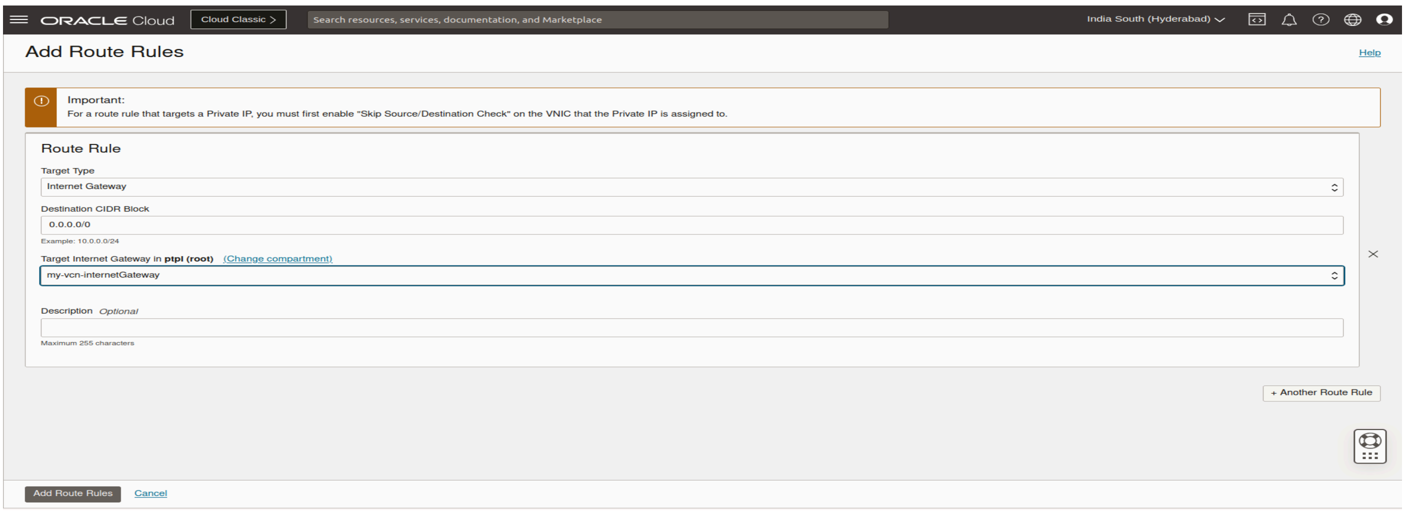
- **After creating the IGW, navigate to the Route Tables section under Networking.**
- **Click on the default route table.**

Add Route Rule:

- **Destination CIDR Block: Enter `0.0.0.0/0` to allow all traffic to the internet.**
- **Target Type: Choose Internet Gateway.**
- **Target: Select the internet gateway (`myIGW`).**
- **Click Save Changes.**

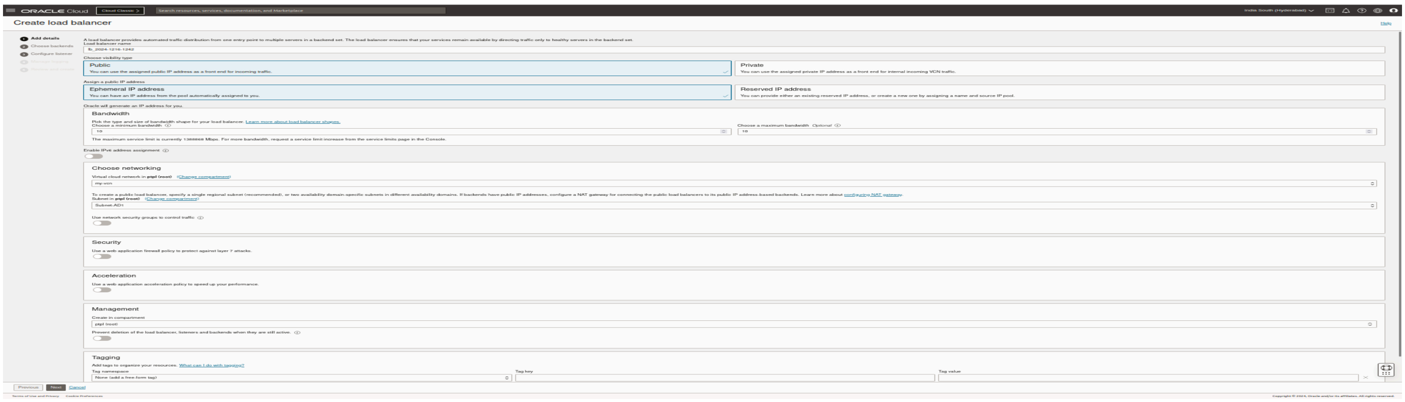






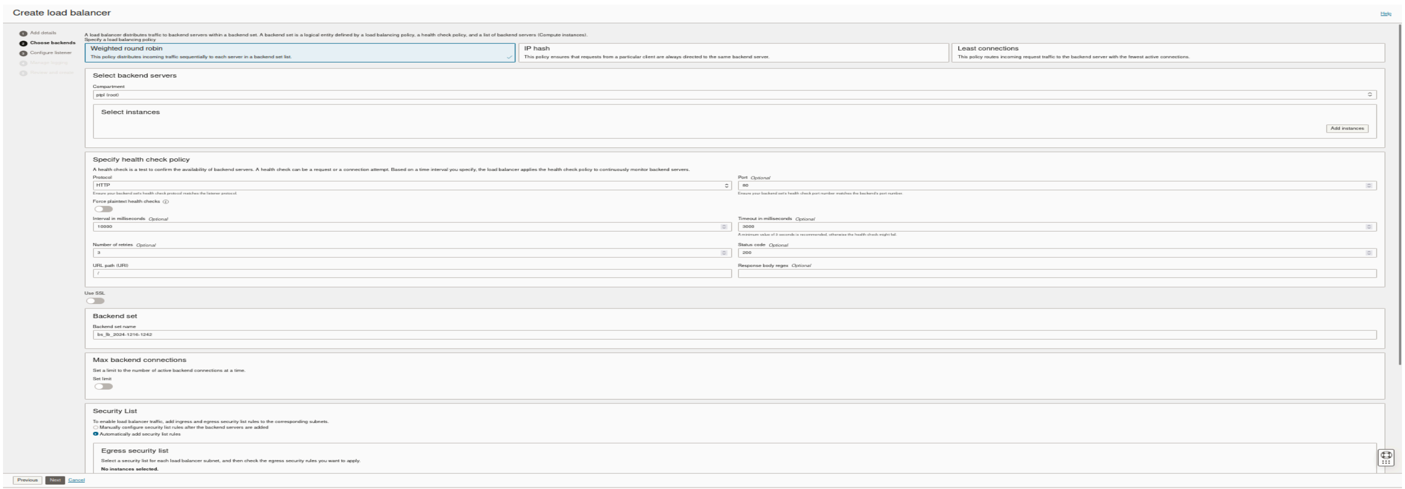
6) Create a Load Balancer (LB)

- **Navigate to Load Balancers**
- **In the Oracle Cloud Console, go to Networking and click Load Balancers.**
- **Click Create Load Balancer.**
- **Name: Enter a name for your Load Balancer, e.g., `myLoadBalancer`**
- **Compartment: Choose the compartment.**
- **Availability Domain: Select the first AD where the LB will be created (e.g., AD-1).**
- **VCN: Select the VCN you created earlier (myVCN).**
- **Subnet: Select the first public subnet (subnet-AD1).**



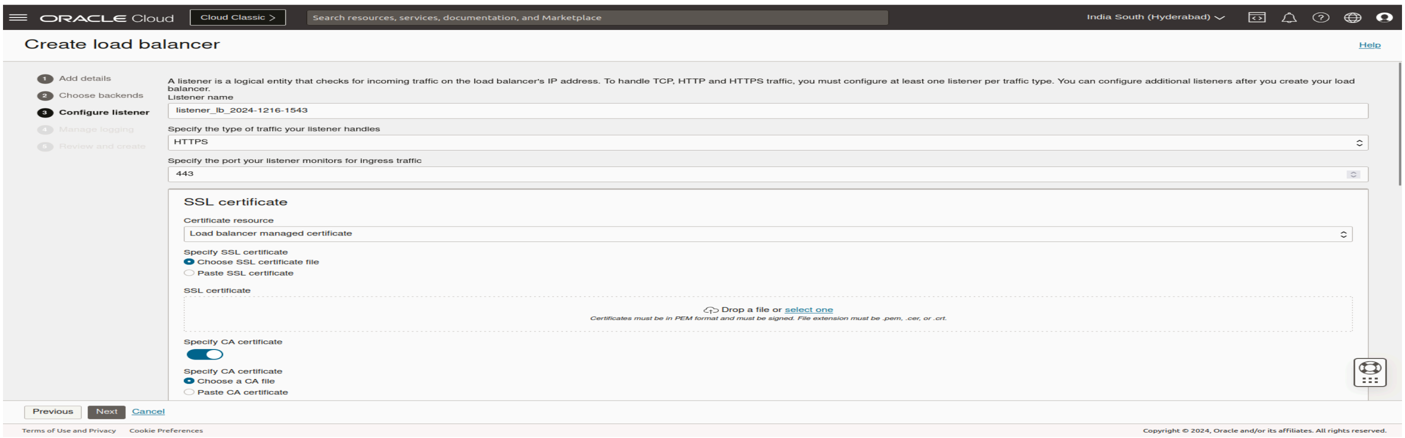
Configure Backend Set

- **Name: Enter a name for the backend set, e.g., `backend-set-1`.**
- **Protocol: Choose the protocol to be used (e.g., HTTP or HTTPS).**
- **Port: Specify the backend port (e.g., `80` for HTTP).**
- **Health Check: Set up the health check:**
- **Type: Select HTTP.**
- **Port: Enter `80` (or the port you configured your backend instances to listen on).**
- **URL Path: Use `/health` or `/` (depending on how your backend instances respond to health checks).**
- **Interval: Set the interval (e.g., `30 seconds`).**
- Keep all other option default and Click **Create**.



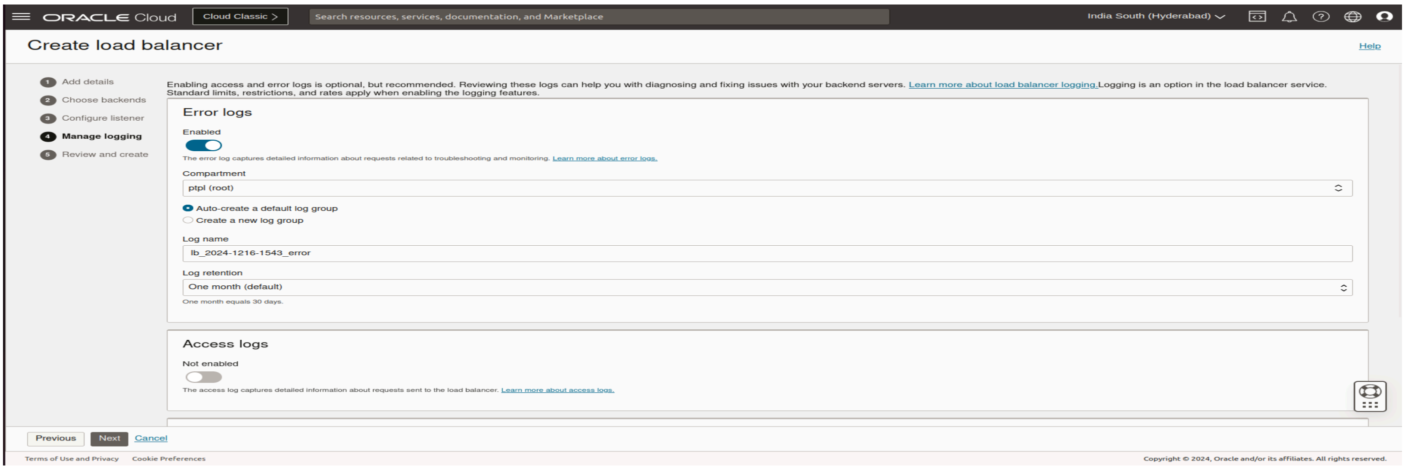
Configure Listener

- **Listener Name: Enter a name, e.g., `http-listener`.**
- **Protocol: Choose HTTP (or HTTPS if using SSL).**
- **Port: Enter `80` (or `443` for HTTPS).**
- **Backend Set: Choose the backend set you created earlier (`backend-set-1`). (If There)**
- **SSL certificate configurations (update as required)**

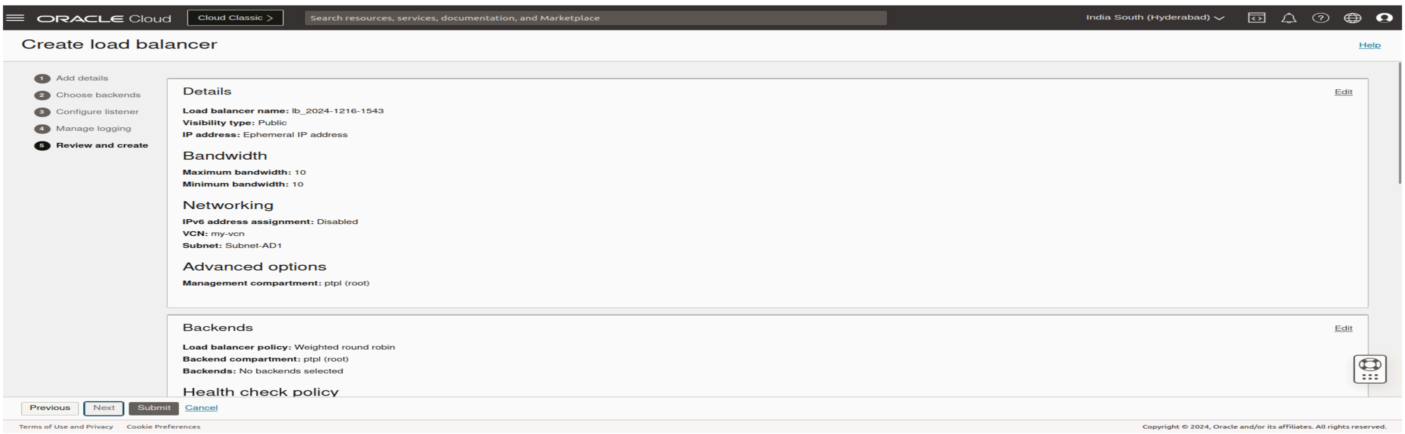


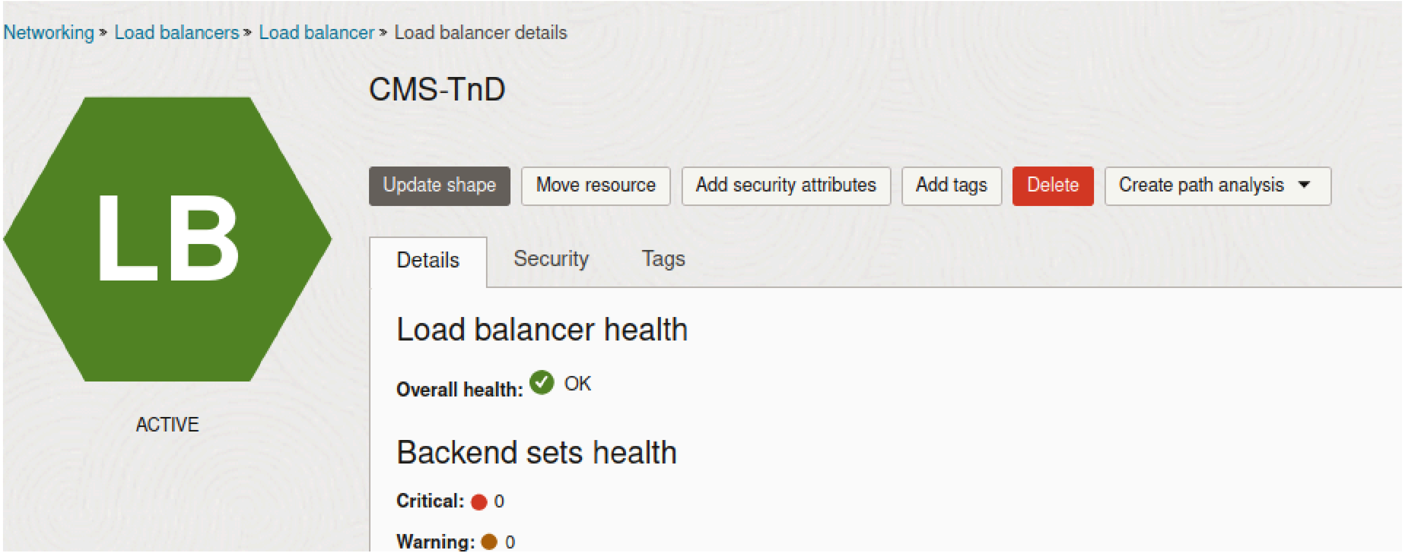
Error Logs

- **Set the configurations as required or keep it default and click next**



Review and Create




Networking » Load balancers » Load balancer » Load balancer details
CMS-TnD
LB
ACTIVE
Update shape
Move resource
Add security attributes
Add tags
Delete
Create path analysis
Details
Security
Tags
Load balancer health
Overall health: OK
Backend sets health
Critical: 0
Warning: 0